

# Nmap - Network Mapper (Student Note)

## NMAP (Network Mapper)

### Learning Objectives

By the end of this module, we should be able to:

- Understand what Nmap is
- Understand hosts, ports, and services
- Perform authorized host discovery
- Scan TCP and UDP ports in a lab
- Understand common port states
- Perform service/version detection
- Understand OS detection
- Use NSE scripts safely
- Analyze Nmap results
- Understand how defenders detect scanning

**Ethical Rule:** We should scan only systems we own or have explicit permission to test. For practical work, we use Kali Linux and an intentionally vulnerable machine such as Metasploitable2.

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## 1. What Is Nmap?

### Short Definition

Nmap (Network Mapper) is a network discovery and security auditing tool used to identify hosts, open ports, running services, and other network characteristics.

Nmap helps us find out:

- Which systems are present on a network
  - Which network “doors” (ports) are reachable
  - What services may be responding behind those doors
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## 2. Why Do We Use Nmap?

Nmap can help us identify:

| Information         | Example                            |
|---------------------|------------------------------------|
| Live Hosts          | Which devices respond              |
| Open Ports          | 22, 80, 443                        |
| Services            | SSH, HTTP, HTTPS                   |
| Versions            | OpenSSH version                    |
| Operating System    | Linux/Windows estimate             |
| Network Information | Reachability and response behavior |

### Example

Target: 192.168.1.10

Port 22 → OPEN → SSH

Port 80 → OPEN → HTTP

Port 443 → OPEN → HTTPS

From this, we learn: the target is exposing remote administration and web services.

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## 3. Basic Networking Concepts Before Nmap

**A. IP Address** — An IP address identifies a device on a network.

Example: 192.168.1.10

Think: **House Address = IP Address**

**B. Port** — A port identifies a network service or communication endpoint.

Examples:

22 → SSH

80 → HTTP

443 → HTTPS

Think: IP Address = Building Port = Door Service = Room / activity behind the door

**C. Service** — A service is an application listening for network communication.

Examples: SSH, HTTP, FTP, DNS, SMTP

**D. Protocol** — Common protocols include: TCP, UDP, ICMP, ARP

Nmap uses different types of network packets and techniques depending on the scan we choose.

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#### 4. Nmap Basic Syntax

`nmap [OPTIONS] TARGET`

Example:

`nmap 192.168.1.10`

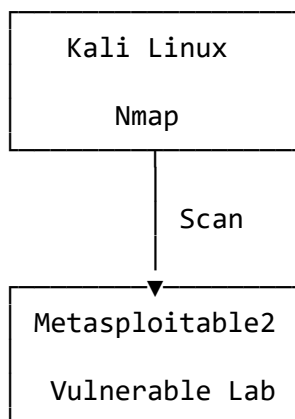
##### Breakdown

`nmap` → Program  
`192.168.1.10` → Target

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### PRACTICAL LAB SETUP

#### Required Machines



First, we find the IP addresses of both machines.

On Kali:

`ip a`

On Metasploitable2:

`ifconfig`

#### Example

Kali IP: 192.168.1.20  
Metasploitable IP: 192.168.1.30

**Tip:** Both machines must be on the same network (same subnet) so they can reach each other. If you're using VirtualBox/VMware, set both VMs to the same "Host-only" or "Internal" network — never "Bridged" on a shared/public network.

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## 5. Our First Nmap Scan

Command:

```
nmap 192.168.1.30
```

Example Output:

Starting Nmap

Nmap scan report for 192.168.1.30

| PORT    | STATE | SERVICE      |
|---------|-------|--------------|
| 22/tcp  | open  | ssh          |
| 80/tcp  | open  | http         |
| 445/tcp | open  | microsoft-ds |

### Understanding the Output

PORT: 22/tcp

Port Number → 22  
Protocol → TCP

STATE: open

Nmap received a response indicating that a service is accepting connections or probes on that port.

SERVICE: ssh

Nmap associates port 22 with the SSH service.

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## 6. Understanding Port States

This is one of the **MOST important concepts** in this module.

### 1. OPEN

Port: 80  
State: OPEN

Meaning: A service appears to be listening and reachable.

TCP SYN  
↓

Target Port 80



SYN-ACK

Nmap may interpret this as: **OPEN**

## 2. CLOSED

Port: 81

State: CLOSED

Meaning: The host responded, but no application is listening on that port.

TCP SYN



Port 81



RST

Nmap interprets this as: **CLOSED**

**Important:** A closed port often means the host itself is reachable — it just doesn't have anything running on that particular port.

## 3. FILTERED

Meaning: Nmap cannot tell whether the port is open, because a firewall or filtering device may be blocking the probe or the response.

Nmap Probe



Firewall



DROP

Result: **FILTERED**

## Quick Summary

| State    | Meaning                                               |
|----------|-------------------------------------------------------|
| Open     | A service is responding                               |
| Closed   | Host responds, but no service is listening            |
| Filtered | A filter (like a firewall) may be blocking visibility |

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## 7. Scanning Specific Ports

Instead of scanning default ports, we can specify exactly which ports to check.

`nmap -p 80 192.168.1.30`

## Meaning

-p → Specify port  
80 → Port number

## Multiple Ports

```
nmap -p 22,80,443 192.168.1.30
```

Nmap checks: 22, 80, 443

## Port Range

```
nmap -p 1-100 192.168.1.30
```

This checks ports 1 through 100.

## All TCP Ports

```
nmap -p- 192.168.1.30
```

This is useful in an authorized lab because some services may be running on non-default ports (ports we wouldn't normally think to check).

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### PRACTICAL ACTIVITY 1

```
nmap <TARGET-IP>
```

Answer:

- Which ports are open?
- Which services are identified?
- Is SSH available?
- Is HTTP available?

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## 8. Host Discovery

Before scanning ports, we can first check whether a host is even reachable — without touching any ports at all.

```
nmap -sn 192.168.1.0/24
```

This is commonly used for authorized network inventory (finding out what's on a network).

### What Does -sn Mean?

-s → Scan  
n → No port scan

Purpose: identify responding hosts without performing a port scan.

Example output:

Nmap scan report for 192.168.1.10  
Host is up

Nmap scan report for 192.168.1.20  
Host is up

Nmap scan report for 192.168.1.30  
Host is up

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## PRACTICAL ACTIVITY 2

Run in lab:

```
nmap -sn 192.168.1.0/24
```

We should identify:

Number of hosts discovered: \_\_\_\_\_

Kali IP: \_\_\_\_\_

Metasploitable IP: \_\_\_\_\_

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## 9. Service and Version Detection

A basic scan may only guess a service based on the “usual” port number (e.g. it assumes port 80 is HTTP). For deeper, more accurate identification, we use:

```
nmap -sV 192.168.1.30
```

-sV

s → Scan

V → Version

Nmap actually sends probes to the service and reads how it responds to identify the exact software and version.

Example:

| PORT   | STATE | SERVICE | VERSION      |
|--------|-------|---------|--------------|
| 22/tcp | open  | ssh     | OpenSSH 4.7  |
| 80/tcp | open  | http    | Apache httpd |

### Why Is Version Detection Important?

Example: Service: SSH – Version: OpenSSH 4.7

A security analyst may then:

- Identify the exact software
  - Check vendor security advisories (public warnings about known bugs)
  - Determine whether the software is outdated
  - Prioritize patching and remediation (fixing the weak spot)
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### PRACTICAL ACTIVITY 3

Run:

```
nmap -sV <TARGET-IP>
```

Create this table:

| Port  | Service | Version |
|-------|---------|---------|
| 22    |         |         |
| 80    |         |         |
| Other |         |         |

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## 10. OS Detection

Nmap can estimate a target's operating system by studying how its network stack behaves — this is called fingerprinting.

```
sudo nmap -O <TARGET-IP>
```

Example output:

OS details: Linux

**Important Note:** OS detection is an *estimate*, not a guarantee. Network devices, firewalls, virtualization, and limited responses can reduce accuracy.

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## 11. Combining Scans

A commonly used combination in an authorized lab is:

```
sudo nmap -sS -sV -O <TARGET-IP>
```

### Breakdown

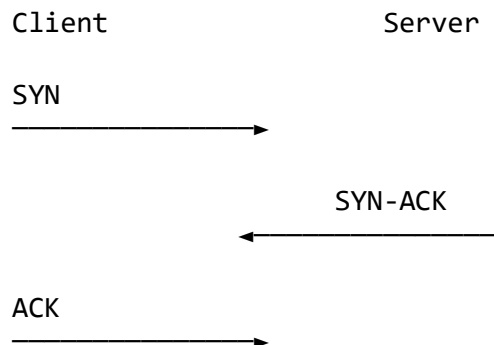
- sS → TCP SYN scan (a common way to check port states)
- sV → Service/version detection
- O → OS detection

This combines several types of information gathering into one command.

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## 12. Understanding TCP SYN Scanning

A normal TCP connection begins with three steps, called the **TCP Three-Way Handshake**:



### How SYN Scanning Works (Simple View)

For an open TCP port:

```
Scanner → SYN      → Target
Target  → SYN-ACK  → Scanner
Scanner → (stops here – does not complete the full connection)
```

The exact packets can be watched directly in a controlled lab using **Wireshark**.

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### PRACTICAL WIRESHARK LAB

1. Start Wireshark on the lab network interface.
  2. Run an authorized scan:  
`sudo nmap -sS <TARGET-IP>`
  3. Use this display filter in Wireshark:  
`tcp`
  4. Students should observe: SYN, SYN-ACK, RST
  5. Compare the packets you see with the port state Nmap reported.
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## 13. UDP Scanning

- **TCP** is connection-oriented (it shakes hands first).
- **UDP** is connectionless (it just sends data, no handshake).

Examples of UDP services:

53 → DNS  
67 → DHCP  
123 → NTP  
161 → SNMP

An authorized UDP scan:

```
sudo nmap -sU -p 53,161 <TARGET-IP>
```

### Why UDP Scanning Can Be Tricky

UDP services often don't reply to probes at all. So "no response" does **not** always mean "closed" — it might just mean the service chose not to answer. This is why Nmap sometimes reports an unclear state such as:

open|filtered

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## 14. Nmap Scan Types (Simple Comparison)

### TCP Connect Scan

```
nmap -sT <TARGET-IP>
```

Concept: uses the operating system's normal, full connection process.

### SYN Scan

```
sudo nmap -sS <TARGET-IP>
```

Concept: uses TCP SYN packets to check port state without completing a full connection.

### UDP Scan

```
sudo nmap -sU <TARGET-IP>
```

Concept: tests UDP ports.

| Scan | Protocol | Basic Purpose                |
|------|----------|------------------------------|
| -sT  | TCP      | Full connection approach     |
| -sS  | TCP      | SYN-based port state testing |
| -sU  | UDP      | UDP port testing             |

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## 15. Nmap Output Formats

Nmap results can be saved to a file so we don't lose them.

### Normal Output

```
nmap -oN scan.txt <TARGET-IP>
```

## XML Output

```
nmap -oX scan.xml <TARGET-IP>
```

## All Major Formats at Once

```
nmap -oA scan <TARGET-IP>
```

This creates multiple output formats using the same base filename (e.g. scan.nmap, scan.xml, scan.gnmap).

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### PRACTICAL ACTIVITY

```
mkdir nmap-results
```

```
cd nmap-results
```

```
nmap -sV -oN lab-scan.txt <TARGET-IP>
```

Check the result:

```
cat lab-scan.txt
```

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## 16. Useful Nmap Options (Cheat Sheet)

| Command  | Purpose                                          |
|----------|--------------------------------------------------|
| nmap IP  | Basic scan                                       |
| -sn      | Host discovery                                   |
| -p 80    | Specific port                                    |
| -p 1-100 | Port range                                       |
| -p-      | All TCP ports                                    |
| -sV      | Service/version detection                        |
| -O       | OS detection                                     |
| -sS      | TCP SYN scan                                     |
| -sT      | TCP Connect scan                                 |
| -sU      | UDP scan                                         |
| -oN      | Save normal output                               |
| -oX      | Save XML output                                  |
| -oA      | Save multiple formats                            |
| -v       | Verbose output (show more detail while scanning) |

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## 17. Nmap Scripting Engine (NSE)

NSE lets Nmap run extra automated checks beyond a plain scan.

```
nmap --script <script-name> <TARGET-IP>
```

For a controlled lab, start with scripts made for discovery and safe information gathering — for example:

```
nmap --script http-title <TARGET-IP>
```

This simply reads the title of a website running on the target, so we get a small taste of what NSE does before going further.

### Simple concept:

```
Nmap
  ↓
NSE Script
  ↓
Additional Probe
  ↓
Additional Information
```

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## 18. Defensive Perspective [?](#)

As a SOC (Security Operations Center) analyst, Nmap scanning may appear in logs like this:

```
One IP
  ↓
Many destination ports
  ↓
Short time period
```

Example:

```
192.168.1.20 → Port 21
192.168.1.20 → Port 22
192.168.1.20 → Port 23
192.168.1.20 → Port 25
192.168.1.20 → Port 80
```

This pattern may indicate: **port scanning or network reconnaissance.**

**Important:** This is an inference, not automatic proof of malicious activity. Always investigate:

- Source IP
- Destination IP
- Time period
- Number of ports
- Connection results
- Whether the scanner was authorized

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## LAB

**Objective:** Perform an authorized scan of Metasploitable2.

### Step 1 — Check IP Address

```
ip a
```

### Step 2 — Basic Scan

```
nmap <TARGET-IP>
```

Record: Open Ports — Services —

### Step 3 — Specific Ports

```
nmap -p 21,22,23,80 <TARGET-IP>
```

### Step 4 — Service Detection

```
nmap -sV <TARGET-IP>
```

### Step 5 — OS Detection

```
sudo nmap -O <TARGET-IP>
```

### Step 6 — Save Results

```
nmap -sV -oN student-scan.txt <TARGET-IP>
```

### Step 7 — Short Write-Up

In a few sentences, answer:

1. What did you find open on the target?
2. Which one service looked most interesting or risky, and why?
3. If you were a defender watching this scan happen, what would you notice in the logs?

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## Quick Recap

- Nmap tells us: is the host alive → what ports are open → what's running on them → what OS it might be
- Port states: **open** (service answering), **closed** (host up, nothing listening), **filtered** (a firewall is in the way)
- -sV gets us versions, -O estimates the OS, -sS/-sT/-sU are our scan types
- Always save results with -oN or -oA — never rely on just reading the terminal
- Only scan machines we own or are authorized to test